Name: _____

WORKSHEET

Derivatives of polynomials

1 Find the derivatives of the following polynomials.

a $3x^2 - 7x + 2$ _____

b $5x^3 + 7x^2 - 4x - 12$ _____

c $x^5 + 3x^3 - 2x^2 + 1$ _____

d $-9x^4 - 8x^3 + 6x^2 - 11x - 4$ _____

e $x^3 + \sqrt{2}\,x^2 - 7x$ _____

f $3x^4 - \frac{1}{5}x^2 + \frac{4}{7}x$ _____

g $5x^8 - 3x^5 + 4x^2 + 16$ _____

h $x^{11} + 7x^7 - 2x^4 + 15x^3 - 43x$ _____

i $2x^9 - 3x^8 - 8x^7 - 4x^6 + 7x^5$ _____

j $0.5x^8 - 0.2x^5 + 0.4x^2 - 0.5x$ _____

2 Find the following.

a $\frac{d}{dx}(3x - 4x^2 + 7x^6)$ _____

b $\frac{d}{dx}(14x^5 - 9x^3 + 12x^4 + 5x^2 - 7 + 2x)$ _____

c $\frac{d}{dx}\left(\frac{x^5}{2} - 3x^2 + \frac{3x^4}{2} - \frac{7}{2}x\right)$ _____

d $\frac{d}{dx}(17x^2 - 2x^6 + 4x^{10} - 7)$ _____

e $\frac{d}{dt}(6t^5 - 3t^8 + 5t^5 - 9t^4 + t + 7t^3)$ _____

f $\frac{d}{dp}(-p^{12} - 3p^4 + 2p^5 - 6p^6)$ _____

g $\frac{d}{dz}\left(-\sqrt{3}z^4 + 0.2z^5 - 7z^3\right)$ _____

h $\frac{d}{dr}(-r^6 + 3.2r^4 - 0.6r^2 + 0.3r)$ _____

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i $\frac{d}{dv}(av^5 + bv^4 + cv^3 + dv^2 + ev + f)$ _____

j $\frac{d}{dt}\left(ut + \frac{1}{2}at^2\right)$ _____

3 Expand and differentiate.

a $(x + 5)(x - 3)$

b $(2x - 5)(3x + 7)$

c $x^2(x + 2)$

d $(x^2 - 1)(x + 6)$

e $(2 - x^2)(4x + 3)$

f $(x - 8)(x^4 + 3x^2)$

g $(x^2 - 5)(x^3 + 2x^2)$

h $(x + 1)(x^4 - 3x^2 - 5x)$

i $(2x + 3)^2$

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j $(x^2 - 2x)^2$

k $(5x - 2x^3)^2$

l $(x^2 - 4x)^2(x + 2)$

4 Find the gradient of $f(x) = x^5 + 5x^4 - 20x^2 - 10x + 2$ for the following values of x.

x	-4	-3	-2	-1	0	1	2
$f'(x)$							

Answers

1 **a** $6x - 7$

b $15x^2 + 14x - 4$

c $5x^4 + 9x^2 - 4x$

d $-36x^3 - 24x^2 + 12x - 11$

e $3x^2 + 2\sqrt{2}\,x - 8x$

f $12x^3 - \frac{2}{5}x + \frac{4}{7}$

g $40x^7 - 15x^4 + 8x$

h $11x^{10} + 49x^6 - 8x^3 + 45x^2 - 43$

i $18x^8 - 24x^7 - 56x^6 - 24x^5 + 35x^4$

j $4x^7 - x^4 + 0.8x - 0.5x$

2 **a** $42x^5 - 8x + 3$

b $70x^4 + 48x^3 - 27x^2 + 10x + 2$

c $\frac{5}{2}x^4 + 6x^3 - 6x - \frac{7}{2}$

d $40x^9 - 12x^5 + 34x$

e $-24t^7 + 25t^4 - 36t^3 + 21t^2 + 12t + 1$

f $-12p^{11} - 36p^5 + 10p^4 - 12p^3$

g $z^4 - 4\sqrt{3}\,z^3 - 21z^2$

h $-6r^5 + 12.8r^3 - 1.2r + 0.3$

i $5av^4 + 4bv^3 + 3cv^2 + 2dv +$

j $u + at$

3 **a** $2x + 2$

b $12x - 1$

c $3x^2 + 4x$

d $3x^2 + 12x - 1$

e $-12x^2 - 6x + 8$

f $5x^4 - 32x^3 + 9x^2 - 48x$

g $5x^4 + 8x^3 - 15x^2 - 20x$

h $5x^4 + 4x^3 - 9x^2 - 16x - 5$

i $8x + 12$

j $4x^3 - 12x^2 - 8x$

k $24x^5 - 80x^3 + 50x$

l $5x^4 - 24x^3 + 64x$

4

x	-4	-3	-2	-1	0	1	2
$f'(x)$	150	-25	-10	15	-10	-25	150

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